

A Short Guide to Proposition C.4

The hybrid Gaussian product bound

Abstract

Proposition C.4 proves the converse bound for the hybrid Gaussian model. It shows that an arbitrary exactly covariant hybrid operation cannot trade classical error against quantum error to beat the product of the separate Gaussian and thermal-state fidelities. The proof uses the residual subinstrument from Lemma C.3, a branchwise stochastic-order estimate from Lemma D.4, and a carefully chosen positive witness. This note develops the required background and explains every step of the product estimate.

1 The problem at the origin

Fix $\gamma > 1$ and a strictly ordered spectrum p . Let $s = d - 1 = \dim \mathbf{H}$ and enumerate the oscillator modes by $\ell = 1, \dots, r_d$. Write

$$q_\ell \in (0, 1), \quad \tau_{q_\ell} = (1 - q_\ell) \sum_{k \geq 0} q_\ell^k |k\rangle\langle k|,$$

and set

$$\Phi_0 = \bigotimes_{\ell=1}^{r_d} \tau_{q_\ell}.$$

At parameter $(h, z) = (0, 0)$, both the hybrid input and the desired target are

$$\Theta_0 = \mathbf{N}_{0, \Sigma_p} \otimes \Phi_0. \tag{1}$$

They are the same state at this one parameter value. Nevertheless, an admissible map in the converse must be covariant with gain $\sqrt{\gamma}$ for *all* translations and displacements. The identity map has the wrong gain and is not an admissible competitor. Amplification necessarily adds noise, which lowers the fidelity even at the origin.

The natural reference operation consists of:

- the classical dilation $x \mapsto \sqrt{\gamma}x$, which sends $\mathbf{N}_{0, \Sigma_p}$ to $\mathbf{N}_{0, \gamma \Sigma_p}$; and
- the product quantum-limited amplifier, which sends τ_{q_ℓ} to $\tau_{q_\ell^{(\gamma)}}$, where

$$q_\ell^{(\gamma)} = 1 - \frac{1 - q_\ell}{\gamma}.$$

Its fidelity with (1) is

$$B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p),$$

where

$$B_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{s/2},$$

$$B_{\text{orb}}(\gamma, p) = \prod_{\ell=1}^{r_d} B(\tau_{q_\ell^{(\gamma)}}, \tau_{q_\ell}).$$

Proposition C.4 proves that even a hybrid operation coupling the classical and quantum components cannot do better.

2 Background I: phase twirling and diagonal states

Let $\hat{N}_\ell$ be the photon-number operator of mode ℓ and

$$U_\theta^{\text{ph}} = \bigotimes_{\ell=1}^{r_d} e^{i\theta_\ell \hat{N}_\ell}.$$

Both the origin input and target are invariant under these independent phase rotations. Given a map Γ , average it as

$$\Gamma^{\text{ph}} = \int \vartheta_{-\theta}^{\text{out}} \circ \Gamma \circ \vartheta_\theta^{\text{in}} d\theta,$$

using normalized Haar measure on the torus. Concavity and unitary invariance of root fidelity imply

$$B(\Gamma^{\text{ph}}(\Theta_0), \Theta_0) \geq B(\Gamma(\Theta_0), \Theta_0).$$

The twirl preserves the required classical translation and quantum displacement covariance. Therefore an upper bound may assume phase covariance from the outset.

If an operator is invariant under every independent phase rotation, it is diagonal in the joint number basis $\{|\mathbf{k}\rangle\langle\mathbf{k}| : \mathbf{k} \in \mathbb{Z}_{\geq 0}^{r_d}\}$. This reduces the final root fidelity to a classical Hellinger sum inside the classical integral.

3 Background II: the reference distributions and witness

Write the target and amplified-reference densities as

$$f_1(y) = \varphi_{0, \Sigma_p}(y), \quad f_\gamma(y) = \varphi_{0, \gamma \Sigma_p}(y).$$

The joint photon-number laws of Φ_0 and of the amplified reference are

$$t_{\mathbf{k}} = \prod_{\ell} (1 - q_\ell) q_\ell^{k_\ell},$$

$$t_{\mathbf{k}}^{(\gamma)} = \prod_{\ell} (1 - q_\ell^{(\gamma)}) (q_\ell^{(\gamma)})^{k_\ell}.$$

Define the square-root likelihood ratios

$$\chi(y) = \sqrt{\frac{f_1(y)}{f_\gamma(y)}}, \quad w_{\mathbf{k}} = \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}}. \quad (2)$$

Because $\gamma > 1$, the Gaussian ratio χ is positive and bounded. The operator

$$W(\hat{\mathbf{N}}) = \sum_{\mathbf{k}} w_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}|$$

is bounded. Moreover,

$$w_{\mathbf{k}} = \prod_{\ell} w_{\ell}(k_{\ell}),$$

where each factor is positive and decreasing because $q_{\ell} < q_{\ell}^{(\gamma)}$. This places W exactly within the scope of Lemma D.4.

The ratios in (2) are chosen so that both reference overlaps factor:

$$\int \frac{f_1(y)}{\chi(y)} dy = \int \sqrt{f_1(y) f_{\gamma}(y)} dy = B_{\text{cl}}(\gamma, d), \quad (3)$$

$$\sum_{\mathbf{k}} \frac{t_{\mathbf{k}}}{w_{\mathbf{k}}} = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = B_{\text{orb}}(\gamma, p). \quad (4)$$

Consequently,

$$\sum_{\mathbf{k}} \int \frac{f_1(y) t_{\mathbf{k}}}{\chi(y) w_{\mathbf{k}}} dy = B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (5)$$

4 Background III: the branchwise conic bound

Lemma D.4 says that if Λ is a completely positive, trace-nonincreasing multimode map with gain- $\sqrt{\gamma}$ displacement covariance and independent phase covariance, then

$$\text{Tr}[\Lambda(\Phi_0) W(\hat{\mathbf{N}})] \leq \text{Tr}[\Lambda(\Phi_0)] \text{Tr}[\tau^{(\gamma)} W(\hat{\mathbf{N}})], \quad (6)$$

where

$$\tau^{(\gamma)} = \bigotimes_{\ell} \tau_{q_{\ell}^{(\gamma)}}.$$

By (4), the final trace in (6) is $B_{\text{orb}}(\gamma, p)$. The estimate is called *conic* because it applies to subnormalized outputs: the right side is scaled by the output trace.

The one-mode result behind this estimate has the usual least-noise stochastic-order interpretation. For a possibly correlated multimode map, however, Lemma D.4 asserts only the product-test inequality (6), not a full multivariate stochastic order. This is sufficient here because W is precisely a product of decreasing one-mode weights.

5 Statement of Proposition C.4

Proposition C.4 (rephrased). *Every exactly covariant completely positive trace-nonincreasing hybrid map Γ satisfies*

$$B(\Gamma(\Theta_0), \Theta_0) \leq B_{\text{univ}}(\gamma, p) := B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \quad (7)$$

Consequently, the flat-prior hybrid Gaussian payoff is at most $B_{\text{univ}}(\gamma, p)$.

In the paper the input and target at the origin are denoted Θ_0^{in} and Θ_0^{tar} . They coincide as in (1); the two labels are retained there to emphasize their different roles in the covariant families.

6 Proof, step by step

6.1 Step 1: phase twirl and apply Lemma C.3

By the phase-twirling argument, assume Γ is phase covariant without decreasing its origin payoff. Lemma C.3 then gives a residual subinstrument $E \mapsto \mathcal{I}_E$. Every branch is displacement covariant with gain $\sqrt{\gamma}$ and phase covariant, so Lemma D.4 applies separately to every $\mathcal{I}_E$.

Define two finite positive measures on the residual classical space:

$$\mu_1(E) = \text{Tr}[\mathcal{I}_E(\Phi_0)], \quad \mu_W(E) = \text{Tr}[\mathcal{I}_E(\Phi_0)W(\hat{\mathbf{N}})]. \quad (8)$$

Applying (6) to the branch $\mathcal{I}_E$ gives, for every Borel set E ,

$$\mu_W(E) \leq B_{\text{orb}}(\gamma, p)\mu_1(E). \quad (9)$$

Thus $\mu_W \leq B_{\text{orb}}\mu_1$ as measures. Since the residual instrument is trace nonincreasing,

$$\mu_1(\mathbf{H}) \leq 1. \quad (10)$$

6.2 Step 2: control a shifted classical Gaussian overlap

The residual representation convolves the ideal dilated Gaussian f_γ with the residual outcome r . Introduce

$$c_\gamma(r) = \int_{\mathbf{H}} \chi(y) f_\gamma(y - r) dy. \quad (11)$$

A direct completion of the square, using the inverse of Σ_p on $\mathbf{H}$, gives

$$c_\gamma(r) = B_{\text{cl}}(\gamma, d) \exp \left[-\frac{\gamma - 1}{2\gamma(\gamma + 1)} \langle r, \Sigma_p^{-1} r \rangle \right]. \quad (12)$$

Therefore

$$0 < c_\gamma(r) \leq c_\gamma(0) = B_{\text{cl}}(\gamma, d). \quad (13)$$

This says that residual classical displacement can only reduce the overlap with the centered target. The maximum occurs when no residual displacement is added.

6.3 Step 3: bound the witness expectation of the actual output

Let

$$\Xi = \Gamma(\Theta_0).$$

The Gaussian convolution in Lemma C.3 makes the classical output absolutely continuous even if the residual measure itself is singular. Phase covariance makes its operator density diagonal:

$$X(y) = \sum_{\mathbf{k}} x_{\mathbf{k}}(y) |\mathbf{k}\rangle \langle \mathbf{k}|.$$

Use the test formula of Lemma C.3 with the classical function χ and the quantum observable W . Equations (8) and (11) give

$$\sum_{\mathbf{k}} \int \chi(y) w_{\mathbf{k}} x_{\mathbf{k}}(y) dy = \int_{\mathbf{H}} c_\gamma(r) \mu_W(dr). \quad (14)$$

Now apply the measure domination (9), the pointwise bound (13), and the mass bound (10):

$$\begin{aligned} \int c_\gamma(r) \mu_W(\mathrm{d}r) &\leq B_{\text{orb}}(\gamma, p) \int c_\gamma(r) \mu_1(\mathrm{d}r) \\ &\leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p). \end{aligned} \quad (15)$$

This is the crucial joint estimate. It shows that classical residual noise and quantum residual noise cannot be correlated in a way that exceeds the product benchmark.

6.4 Step 4: convert the witness bounds into fidelity

By the hybrid fidelity formula from Lemma C.1 and diagonality,

$$B(\Xi, \Theta_0) = \sum_{\mathbf{k}} \int \sqrt{x_{\mathbf{k}}(y) f_1(y) t_{\mathbf{k}}} \mathrm{d}y. \quad (16)$$

For positive u, v, A , the arithmetic–geometric mean inequality gives

$$\sqrt{uv} \leq \frac{1}{2} \left(Au + \frac{v}{A} \right). \quad (17)$$

Apply it pointwise with

$$u = x_{\mathbf{k}}(y), \quad v = f_1(y) t_{\mathbf{k}}, \quad A = \chi(y) w_{\mathbf{k}}.$$

After summing and integrating, the first term is bounded by (15); the second is exactly (5). Hence

$$\begin{aligned} B(\Xi, \Theta_0) &\leq \frac{1}{2} [B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p) + B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p)] \\ &= B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p), \end{aligned}$$

which proves (7).

Finally, Lemma C.2 says that the large-cube average payoff of arbitrary hybrid channels is bounded by the best origin payoff among exactly covariant trace-nonincreasing maps. Applying (7) to that limiting map gives the flat-prior upper bound.

7 Why this particular witness works

The choice $A = \chi w$ in (17) is not arbitrary. Equality in (17) occurs when $Au = v/A$, or

$$A = \sqrt{v/u}.$$

For the reference output

$$u = f_\gamma(y) t_{\mathbf{k}}^{(\gamma)}, \quad v = f_1(y) t_{\mathbf{k}},$$

this optimal ratio is exactly

$$\sqrt{\frac{f_1(y) t_{\mathbf{k}}}{f_\gamma(y) t_{\mathbf{k}}^{(\gamma)}}} = \chi(y) w_{\mathbf{k}}.$$

Thus the witness is the tangent functional to root fidelity at the product quantum-limited output. The rest of the proof establishes that every admissible output lies on the correct side of this tangent hyperplane.

8 Dependencies and logical role

The proof uses the following results.

- Lemma C.1 supplies the integral formula for fidelity of hybrid operator-valued measures.
- Lemma C.3 represents an exactly covariant hybrid map by a residual subinstrument and gives the Gaussian convolution formula.
- Lemma D.4 gives the branchwise multimode conic stochastic-order estimate. Its proof tensorizes the one-mode estimate in Lemma D.3, which in turn uses the constant-trace reduction in Lemma D.1 and the least-noise theorem for Gaussian amplifiers.
- Joint concavity and unitary invariance justify phase twirling.
- A Gaussian completion of the square and the elementary inequality (17) finish the estimate.

Proposition C.4 is the final analytic step in the upper bound for the hybrid Gaussian flat-prior theorem. Lemma C.2 reduces arbitrary channels to an exactly covariant limiting map, and C.4 bounds that map by B_{univ} . The explicit product of the classical and orbital factors is therefore a consequence of the witness argument, not an assumption that the competing channel itself factors.

9 Minimal prerequisite checklist

To follow the numerical part of the proof, it is enough to know Gaussian densities, geometric photon-number laws, root fidelity for commuting states, and the arithmetic–geometric mean inequality. The structural reduction also requires phase twirling, covariant completely positive maps, quantum subinstruments, and the stochastic-order estimate in Lemma D.4. The central idea is that the square-root likelihood ratio of the product reference state simultaneously tests the classical and quantum components and rules out any advantage from coupling them.